



Coláiste na Tríonóide, Baile Átha Cliath
Trinity College Dublin

Ollscoil Átha Cliath | The University of Dublin

Faculty of Engineering, Mathematics and Science

School of Computer Science & Statistics

Integrated Computer Science
Integrated Engineering
Computer Science and Business
Computer Science and Language

Semester 1 2023

Year 4

Computer Vision

13th December 2023

RDS-Sim Court

09.30-11.30

Dr Kenneth Dawson-Howe

Instructions to Candidates:

Attempt two questions. All questions carry equal marks. Each question is scored out of a total of 50 marks.

For Application questions, a series of computer vision operations must be detailed to solve the application problem. The input to and output from each technique used must be clearly and precisely stated. How each technique is used within the context of the application must also be described including the setting of any parameters.

For Compare and Contrast type questions marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques.

In all questions you must describe computer vision theory and should not refer to code or library calls (OpenCV or any other library).

1. (a) **[APPLICATION QUESTION]** Describe how you would find blue signs (such as those in the pictures below) and recognise the writing and arrow signs. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[30 marks]



1. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- Canny edge detection.
- Connected Components Analysis.
- Mean Shift Segmentation.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[20 marks]

2. (a) **[APPLICATION QUESTION]** Describe how you would track the two players on a tennis court (who will change ends regularly and will also go out of view from time to time), from a video of the court (from an unmoving camera) with views such as those shown below. You may **not** assume that the clothing will be as shown. The players can wear any clothing they wish to wear. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[25 marks]



2. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- Support Vector Machines.
- Robust Object Matching using a cascade of Haar classifiers.
- Scale Invariant Feature Transform.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]

3. (a) **[APPLICATION QUESTION]** By finding the *vanishing lines* (as shown in red below) describe how you would obtain a front view of a building. Your solution must consist of a series of computer vision techniques and you must provide details of how the techniques will be applied including expected input and output for each technique.

[25 marks]



Original image



Front view



The *vanishing lines* (which correspond to lines which are parallel in the real world) intersect in a single point (called the vanishing point; shown in green).

3. (b) **[COMPARE & CONTRAST QUESTION]** Compare and contrast:

- Hough for circles.
- Chamfer matching for finding circles.
- Mean shift segmentation followed by an evaluation of a measure of circularity.

You must provide a list of the differences and similarities between the techniques. Each of the differences and similarities must be clearly explained. **NOTE:** Marks will only be awarded for the detailed comparison of techniques. No marks will be awarded for separate descriptions of the techniques

[25 marks]